

Introduction

Motivation and Context

The rapid proliferation of multiagent AI systems has created novel attack surfaces that traditional cybersecurity frameworks were not designed to address. Unlike monolithic applications where security boundaries are well-defined, multiagent architectures introduce inter-agent communication channels, delegated authority chains, and emergent collective behaviors that adversaries can exploit. When an AI agent can persuade, instruct, or deceive another agent, the attack surface shifts from code vulnerabilities to cognitive manipulation—a fundamentally different threat model requiring fundamentally different defenses.

Industry adoption of multiagent architectures has accelerated dramatically. By late 2025, McKinsey found that 23% of organizations were scaling agentic AI in some part of their enterprise, with an additional 39% actively experimenting [?]. Gartner projects that 40% of enterprise applications will incorporate task-specific AI agents by the end of 2026, up from under 5% in 2025, with 70% of enterprises deploying agentic AI in IT infrastructure operations by 2029 [?]. Enterprise deployments

COGNITIVE INTEGRITY FRAMEWORK (CIF)

Layered Defense Architecture for Multiagent AI Systems



Generated by: arch/evaluation/figures/cif_comprehensive.py

delta^d exponential decay), Byzantine Consensus (semantic BFT for collective decisions), and Provenance Attestation (cryptographic message origin tracking). Arrows indicate information flow, with

Relationship to Paper Series

This paper assumes familiarity with the formal framework developed in Part 1, particularly:

- ▶ **Trust Calculus** (Section 3 and 4 of Part 1): Bounded delegation with δ^d decay
- ▶ **Defense Composition Algebra** (Section 5 of Part 1): Series and parallel composition theorems
- ▶ **Integrity Properties** (Section 7 of Part 1): Belief consistency, goal preservation, trust boundedness

All notation follows the canonical reference in Part 1 Appendix (sec:notation-reference). For practical deployment guidance including checklists and operational considerations, see Part 3.

Paper Organization

The remainder of this paper is structured as follows:

sec:related-work: Related Work positions CIF relative to prompt injection defenses, Byzantine fault tolerance, trust systems, and multiagent safety research.

sec:methodology: Methodology: Implementation Details describes the architectural realization of CIF and presents pseudocode for each of the six defense mechanisms.

sec:attack-corpus: Attack Corpus describes the 950-attack evaluation dataset with examples and generation methodology.

sec:results: Experimental Validation details the experimental setup, six target architectures, evaluation protocol, and key findings.

sec:extended-results: Extended Results provides per-architecture breakdowns, statistical significance testing, sensitivity analysis, ablation studies, and scalability benchmarks.